

Contrast Limit Adaptive
Histogram Equalization
(CLAHE)


```
graph TD; A([Contrast Limit Adaptive Histogram Equalization (CLAHE)]) --> B[/Read Image/]; B --> C[Apply Mean and Median Filter to Image]; C --> D[Find frequency counts per pixel value]; D --> E[Determine probability for each occurrence using probability function]; E --> F[Calculate cumulative distribution probability per pixel value]; F --> G[Equalization mapping of all pixels]; G --> H[/Display processed image/]; H --> I([Return]);
```

The flowchart illustrates the CLAHE process. It begins with an oval node labeled 'Contrast Limit Adaptive Histogram Equalization (CLAHE)'. An arrow points down to a parallelogram input node 'Read Image'. This is followed by a sequence of rectangular process nodes: 'Apply Mean and Median Filter to Image', 'Find frequency counts per pixel value', 'Determine probability for each occurrence using probability function', 'Calculate cumulative distribution probability per pixel value', and 'Equalization mapping of all pixels'. The next step is a parallelogram output node 'Display processed image', followed by a final oval node 'Return'. All steps are connected by downward-pointing arrows.

Read Image

Apply Mean and Median
Filter to Image

Find frequency counts per
pixel value

Determine probability for
each occurrence using
probability function

Calculate cumulative
distribution probability per
pixel value

Equalization mapping of all
pixels

Display processed image

Return